

# Relations

## Relations and Their properties

**Defintion 1:** Let  $A$  and  $B$  be sets. A *binary relation from  $A$  to  $B$*  is a subset of  $A \times B$ .

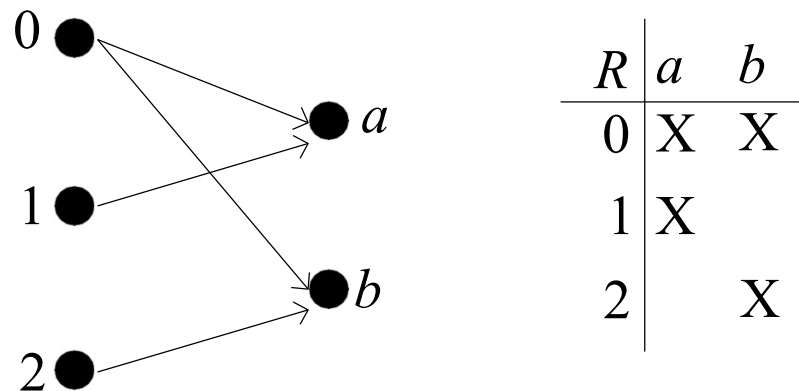
A binary relation from  $A$  to  $B$  is a set  $R$  of ordered pairs where the first element of each ordered pair comes from  $A$  and the second element comes from  $B$ .

We use the notation  $aRb$  to denote that  $(a, b) \in R$ .

When  $(a, b)$  belongs to  $R$ ,  $a$  is said to be **related** to  $b$  by  $R$ .

**Example:** Let  $A = \{0, 1, 2\}$  and  $B = \{a, b\}$ .  
 Then  $R = \{(0, a), (0, b), (1, a), (2, b)\}$  is a relation from  $A$  to  $B$ .  
 This means that  $0Ra$ .

Relations can be represented graphically, as shown in Figure 1, using arrows to represent ordered pairs. Another way to represent this relation is to use a table, which is also done in Figure 1.



**Figure:** Displaying the Ordered Pairs in the Relation  $R$  from Example 3.

## **Functions as Relations**

A function  $f$  from a set  $A$  to a set  $B$  assigns exactly one element of  $B$  to each element of  $A$ . The graph of  $f$  is the set of ordered pairs  $(a, b)$  such that  $b = f(a)$ . Since the graph of  $f$  is a subset of  $A \times B$ , it is a relation from  $A$  to  $B$ . Moreover, the graph of a function has the property that every element of  $A$  is the first element of exactly one ordered pair of the graph.

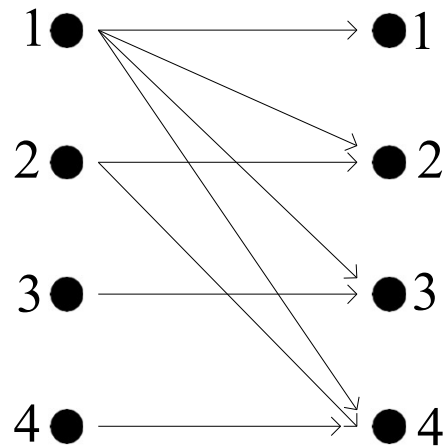
## Relations on a Set

**Definition 2:** A *relation* on the set  $A$  is a relation from  $A$  to  $A$ .

**Example:** Let  $A$  be the set  $\{1, 2, 3, 4\}$ . Which ordered pairs are in the relation  $R = \{(a, b) \mid a \text{ divides } b\}$ ?

Solution:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$



|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | × | × | × | × |
| 2 |   | × |   | × |
| 3 |   |   | × |   |
| 4 |   |   |   | × |

**Example:** Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs  $(1, 1)$ ,  $(1, 2)$ ,  $(2, 1)$ ,  $(1, -1)$ , and  $(2, 2)$ ?

Solution:

$(1, 1)$  is in  $R_1, R_3, R_4, R_6$

$(1, 2)$  is in  $R_1, R_6$

$(2, 1)$  is in  $R_2, R_5, R_6$

$(1, -1)$  is in  $R_2, R_3, R_6$

$(2, 2)$  is in  $R_1, R_3, R_4$

## Properties of Relations on sets

**Definition 3:** A relation  $R$  on a set  $A$  is called *reflexive* if  $(a, a) \in R$  for every element  $a \in A$ .

The relation  $R$  on the set  $A$  is reflexive if  $\forall a \in A ((a, a) \in R)$ .

A relation on  $A$  is reflexive if every element of  $A$  is related to itself.

**Example:** Consider the following relations on  $\{1, 2, 3, 4\}$ :

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Which of these relations are reflexive?

Solution:

The following relations are reflexive:

$$R_3 = \{(\mathbf{1, 1}), (1, 2), (1, 4), (2, 1), (\mathbf{2, 2}), (\mathbf{3, 3}), (4, 1), (\mathbf{4, 4})\}$$

$$R_5 = \{(\mathbf{1, 1}), (1, 2), (1, 3), (1, 4), (\mathbf{2, 2}), (2, 3), (2, 4), (\mathbf{3, 3}), (3, 4), (\mathbf{4, 4})\}$$

**Example:** Which of the following relations on integers are reflexive?

$$\begin{aligned} R_1 &= \{(a, b) \mid a \leq b\} \\ R_2 &= \{(a, b) \mid a > b\} \\ R_3 &= \{(a, b) \mid a = b \text{ or } a = -b\} \\ R_4 &= \{(a, b) \mid a = b\} \\ R_5 &= \{(a, b) \mid a = b + 1\} \\ R_6 &= \{(a, b) \mid a + b \leq 3\} \end{aligned}$$

Solution:

The following relations are reflexive:

$$\begin{aligned} R_1 &= \{(a, b) \mid a \leq b\} \text{ (since } a \leq a \text{ for every integer } a) \\ R_3 &= \{(a, b) \mid a = b \text{ or } a = -b\} \text{ (since } a = a \text{ for every integer } a) \\ R_4 &= \{(a, b) \mid a = b\} \text{ (since } a = a \text{ for every integer } a) \end{aligned}$$

**Example:** Is the “divides” relation on the set of positive integers reflexive?

Solution:

Since  $a \mid a$  whenever  $a$  is a positive integer, the “divides” relation is reflexive.



**Definition 4:** A relation  $R$  on a set  $A$  is called *symmetric* if and only if  $(b, a) \in R$  whenever  $(a, b) \in R$ , for all  $a, b \in A$ .

The relation  $R$  on the set  $A$  is symmetric if

$$\forall a \in A \forall b \in A ((a, b) \in R \leftrightarrow (b, a) \in R).$$

**Example:** Which of the following relations on  $\{1, 2, 3, 4\}$  are symmetric?

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Solution:

The following relations are symmetric, because in each case  $(b, a)$  belongs to the relation whenever  $(a, b)$  does:

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

**Example:** Which of the following relations on integers are symmetric?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Solution:

The following relations are symmetric:

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} \text{ [If } a = b, \text{ then } b = a. \text{ If } a = -b \text{ then, } b = -a]$$

$$R_4 = \{(a, b) \mid a = b\} \text{ [If } a = b \text{ then, } b = a.]$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} \text{ [If } a + b \leq 3 \text{ then, } b + a \leq 3]$$

The following relations are not symmetric:

$$R_1 = \{(a, b) \mid a \leq b\} \text{ [If } a \leq b \text{ then, } b \not\leq a]$$

$$R_2 = \{(a, b) \mid a > b\} \text{ [If } a > b \text{ then, } b \not> a]$$

$$R_5 = \{(a, b) \mid a = b + 1\} \text{ [If } a = b + 1 \text{ then, } b \neq a + 1]$$

**Example:** Is the “divides” relation on the set of positive integers symmetric?

Solution:

This relation is not symmetric, since  $1 \mid 2$ , but  $2 \nmid 1$  .

**Definition 5:** A relation  $R$  on a set  $A$  is called *transitive* if whenever  $(a, b) \in R$  and  $(b, c) \in R$ , then  $(a, c) \in R$ , for all  $a, b, c \in A$ .

The relation  $R$  on a set  $A$  is transitive if we have

$$\forall a \in A \forall b \in A \forall c \in A ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$$

**Example:** Which of the following relations on  $\{1, 2, 3, 4\}$  are transitive?

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Solution:

The following relation is transitive:

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$[(3, 2) \text{ and } (2, 1) \rightarrow (3, 1)$$

$$(4, 2) \text{ and } (2, 1) \rightarrow (4, 1)$$

$$(4, 3) \text{ and } (3, 1) \rightarrow (4, 1)$$

$$(4, 3) \text{ and } (3, 2) \rightarrow (4, 2)]$$

**Example (contd.):**

The following relation is transitive:

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$[(1, 1) \text{ and } (1, 2) \rightarrow (1, 2)$$

$$(1, 1) \text{ and } (1, 3) \rightarrow (1, 3)$$

$$(1, 1) \text{ and } (1, 4) \rightarrow (1, 4)$$

$$(1, 2) \text{ and } (2, 2) \rightarrow (1, 2)$$

$$(1, 2) \text{ and } (2, 3) \rightarrow (1, 3)$$

$$(1, 2) \text{ and } (2, 4) \rightarrow (1, 4)$$

$$(1, 3) \text{ and } (3, 3) \rightarrow (1, 3)$$

$$(1, 3) \text{ and } (3, 4) \rightarrow (1, 4)$$

$$(1, 4) \text{ and } (4, 4) \rightarrow (1, 4)$$

$$(2, 2) \text{ and } (2, 3) \rightarrow (2, 3)$$

$$(2, 2) \text{ and } (2, 4) \rightarrow (2, 4)$$

$$(2, 3) \text{ and } (3, 3) \rightarrow (2, 3)$$

$$(2, 3) \text{ and } (3, 4) \rightarrow (2, 4)$$

$$(2, 4) \text{ and } (4, 4) \rightarrow (2, 4)$$

$$(3, 3) \text{ and } (3, 4) \rightarrow (3, 4)$$

$$(3, 4) \text{ and } (4, 4) \rightarrow (3, 4)]$$

**Example (contd.):**

The following relation is transitive:

$$R_6 = \{(3, 4)\}$$

[There is no pair like  $(4, c)$  which would imply the existence of the pair  $(3, c)$ ]

The following relations are not transitive:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

[There are pairs  $(3, 4)$  and  $(4, 1)$  but not  $(3, 1)$ ]

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

[There are pairs  $(2, 1)$  and  $(1, 2)$  but not  $(2, 2)$ ]

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

[There are pairs  $(4, 1)$  and  $(1, 2)$  but not  $(4, 2)$ ]

**Example:** Which of the following relations on integers are transitive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Solution:

The following relations are transitive:

$$R_1 = \{(a, b) \mid a \leq b\} [(a \leq b) \text{ and } (b \leq c) \rightarrow (a \leq c)]$$

$$R_2 = \{(a, b) \mid a > b\} [(a > b) \text{ and } (b > c) \rightarrow (a > c)]$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} [(a = \pm b) \text{ and } (b = \pm c) \rightarrow (a = \pm c)]$$

$$R_4 = \{(a, b) \mid a = b\} [(a = b) \text{ and } (b = c) \rightarrow (a = c)]$$

The following relations are not transitive:

$$R_5 = \{(a, b) \mid a = b + 1\} [(2, 1), (1, 0) \in R_5 \text{ but } (2, 0) \notin R_5]$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} [(2, 1), (1, 2) \in R_6 \text{ but } (2, 2) \notin R_6]$$

**Example:** Is the “divides” relation on the set of positive integers transitive?

Solution:

If  $a$  divides  $b$ , then  $b = ak$ , where  $k$  is a positive integer

If  $b$  divides  $c$ , then  $c = bl$ , where  $l$  is a positive integer

Then  $c = bl = akl = a(kl)$ , where  $kl$  is a positive integer. Therefore,  $a$  divides  $c$

Hence, the “divides” relation is transitive.